

Building LLMs from the Ground Up: A 3-hour Coding Workshop



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445



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If you'd like to spend a few hours this weekend to dive into Large Language Models (LLMs) and understand how they work, I've prepared a 3-hour coding workshop presentation on implementing, training, and using LLMs.



Building LLMs from the Ground Up: A 3-hour Coding Workshop

Sebastian Raschka



前往平台观看：

Below, you'll find a table of contents to get an idea of what this video covers (the video itself has clickable chapter marks, allowing you to jump directly to topics of interest):

0:00 – Workshop overview

2:17 – Part 1: Intro to LLMs

9:14 – Workshop materials

10:48 – Part 2: Understanding LLM input data

23:25 – A simple tokenizer class

41:03 – Part 3: Coding an LLM architecture

45:01 – GPT-2 and Llama 2

1:07:11 – Part 4: Pretraining

1:29:37 – Part 5.1: Loading pretrained weights

1:45:12 – Part 5.2: Pretrained weights via LitGPT

1:53:09 – Part 6.1: Instruction finetuning

2:08:21 – Part 6.2: Instruction finetuning via LitGPT

02:26:45 – Part 6.3: Benchmark evaluation

02:36:55 – Part 6.4: Evaluating conversational performance

02:42:40 – Conclusion

It's a slight departure from my usual text-based content, but the last time I did this a few months ago, it was so well-received that I thought it might be nice to do another one!

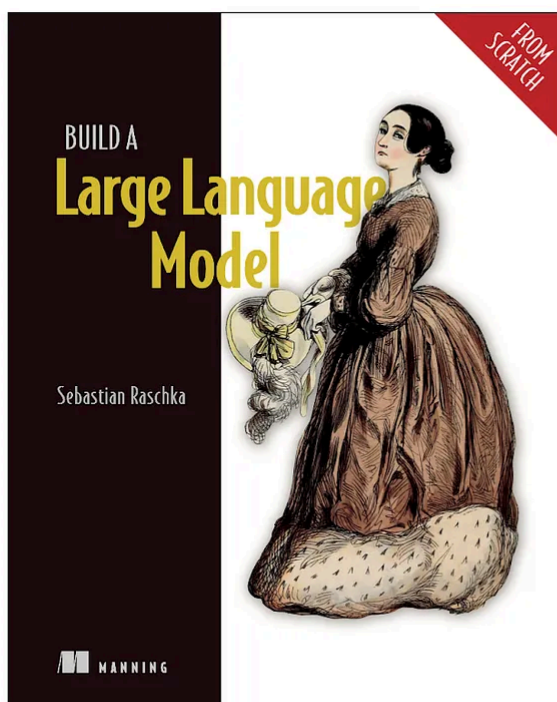
Happy viewing!

References

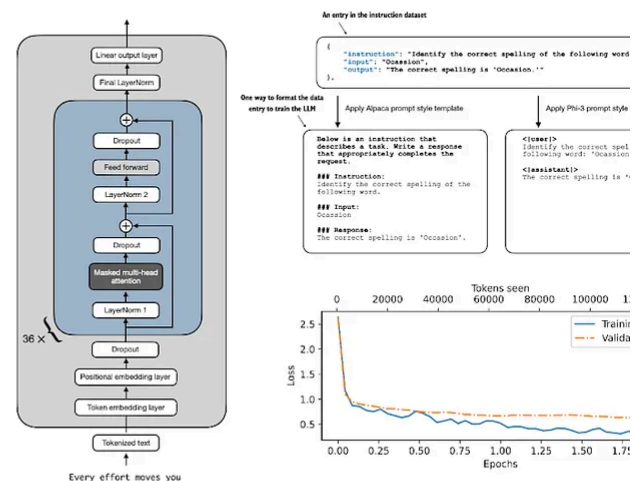
1. [Build an LLM from Scratch book](#)
2. [Build an LLM from Scratch GitHub repository](#)
3. [GitHub repository with workshop code](#)

4. [Lightning Studio for this workshop](#)
5. [LitGPT GitHub repository](#)

This magazine is a personal passion project. For those who wish to support me, please consider purchasing a copy of my [Build a Large Language Model \(From Scratch\)](#) because I am confident that you'll get lots out of this book as it explains how LLMs work in a level of detail that is not found anywhere else.)



Deeply understand LLMs by implementing them from the ground up



Build a Large Language Model (From Scratch) now [available on Amazon](#)

If you read the book and have a few minutes to spare, I'd really appreciate a [brief review](#). It helps us authors a lot!

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WvG  WvG 2024年8月31日

♥ Liked by Sebastian Raschka, PhD

THANK YOU so much for doing this! I have bought the early release of your book via MEAP a fantastic. Highly recommended for everybody who wants to be hands on and really get a de understanding and appreciation regarding LLMs.

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1 reply by Sebastian Raschka, PhD



Logan Thorneloe  AI for Software Engineers 2024年9月5日

♥ Liked by Sebastian Raschka, PhD

Thank you for the work you put into these! Do you mind if I add this to the machine learning I've put together as a guide to learn LLMs?

♥ LIKE (4)  REPLY

1 reply by Sebastian Raschka, PhD

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